



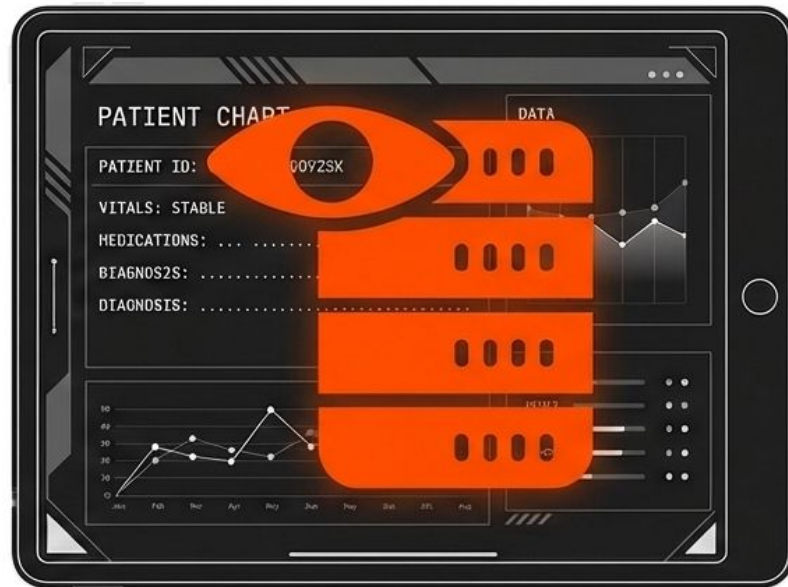
# BREAKING THE BLACK BOX

Architectural Foundations of  
zkLLM

# THE LEGAL RISK

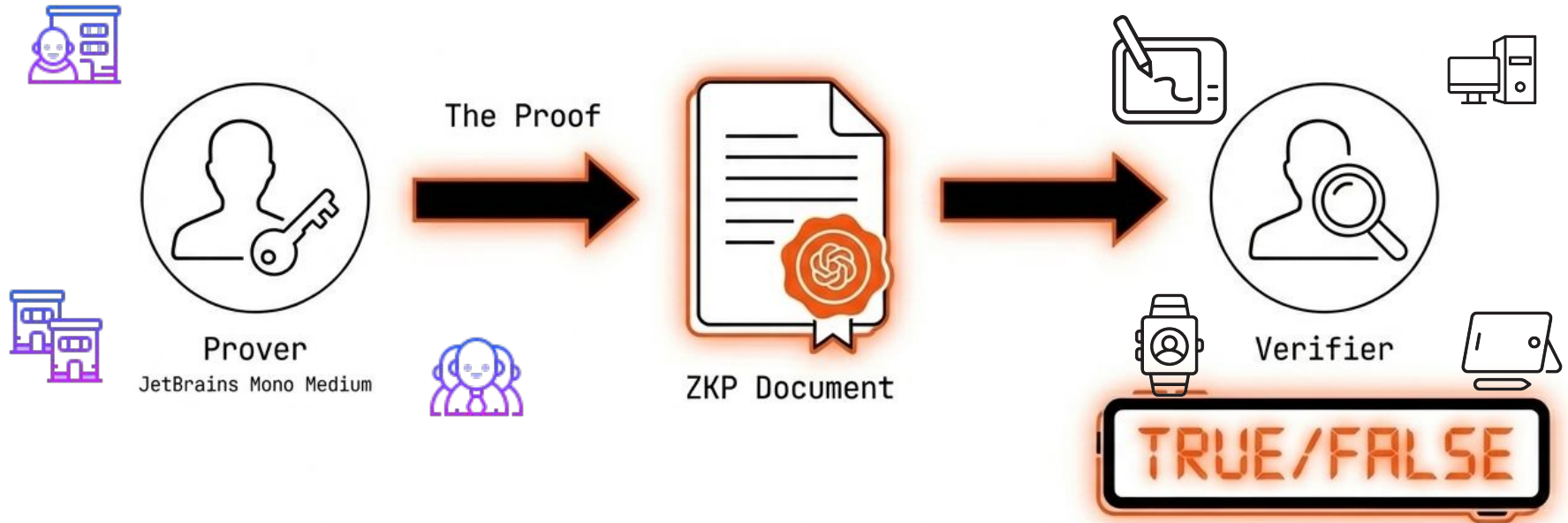


# THE SAFETY RISK



**THE DILEMMA:** How do we audit the process without exposing the data?

# ZERO-KNOWLEDGE BASICS



**COMPLETENESS:** If true,  
the verifier is convinced.

**SOUNDNESS:** If false,  
the prover cannot lie.

**ZERO-KNOWLEDGE:** No  
secret is revealed.

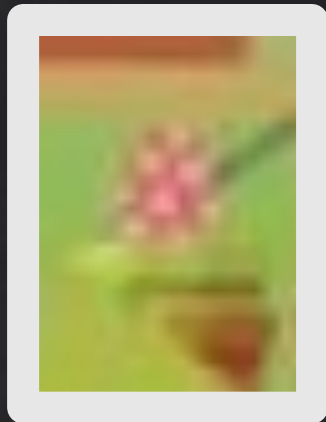
**Example**

**Find The  
Easter Egg?**





# ZK Proof



Do you know where  
it is?

Did I prove I know  
where it is without  
giving any  
information?

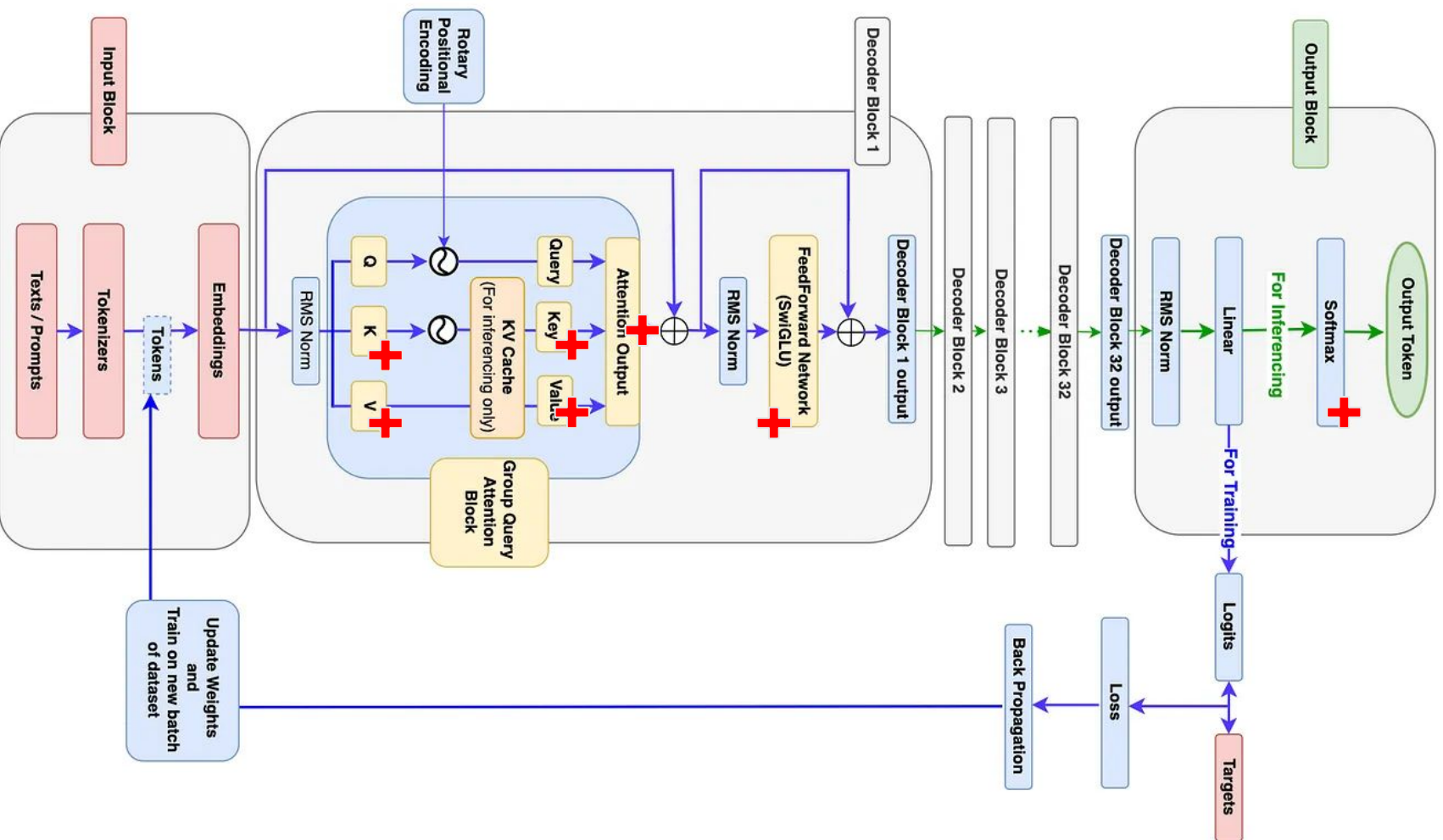


# Zero-Knowledge Proofs guarantee cryptographic truth without data exposure



"I can prove to you with 100% mathematical certainty that I possess a secret, without ever revealing the secret itself."







# ZK Proof

## Step 1: Input



User provides private data.

## Step 2: Process



The hidden AI model runs the inference.

## Step 3: Output Generation

The Prediction (e.g., "Diagnosis: Healthy")

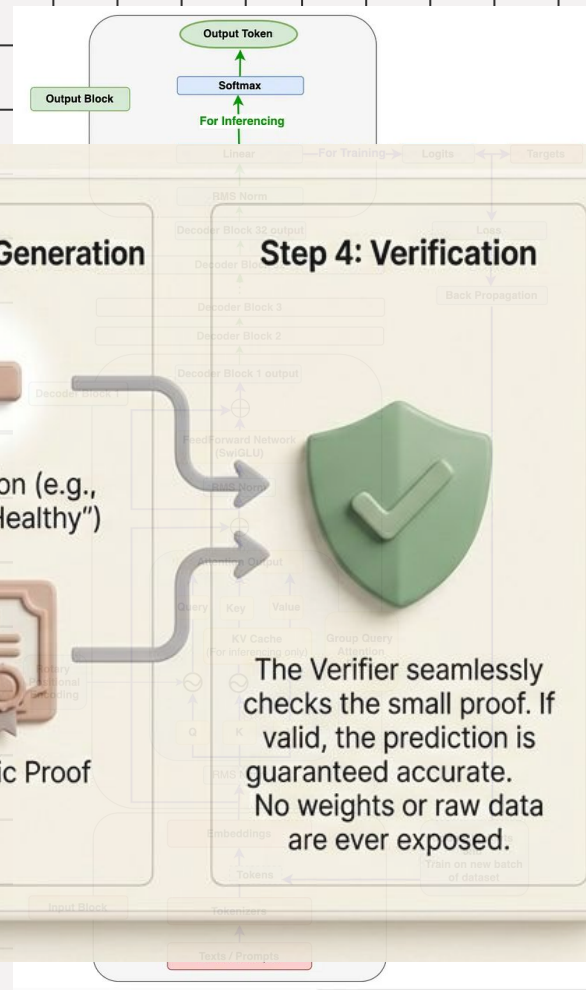


Cryptographic Proof

## Step 4: Verification



The Verifier seamlessly checks the small proof. If valid, the prediction is guaranteed accurate. No weights or raw data are ever exposed.



## The Good: Linear Algebra



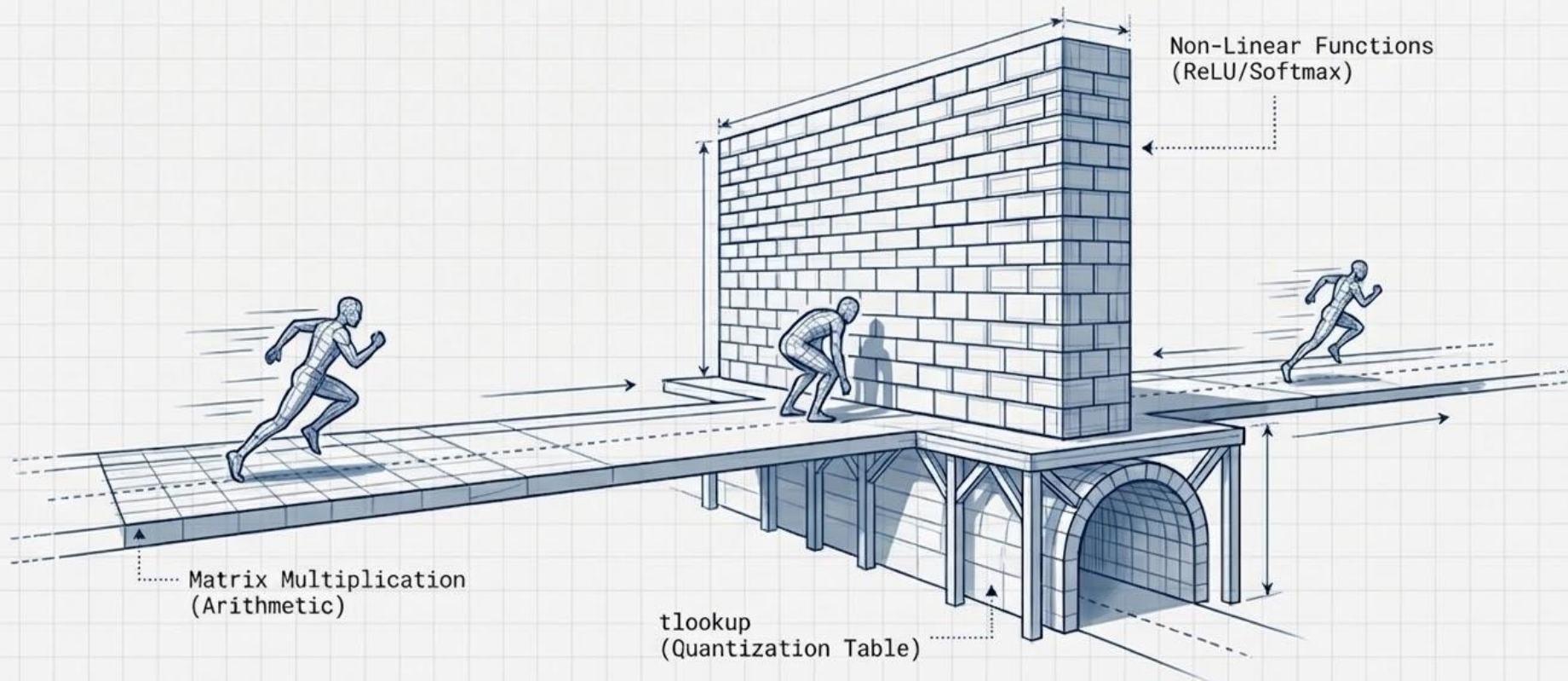
## The Bad: Non-Linear Functions



Large Language Models rely on “non-arithmetic” operations.

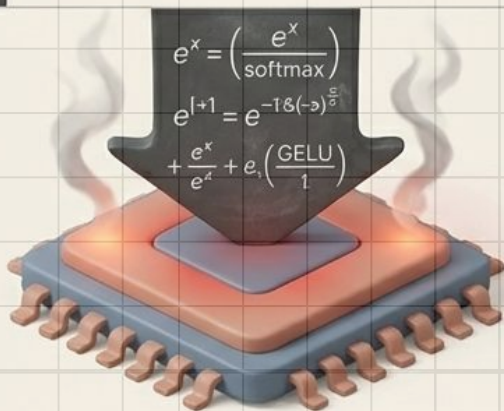
Cryptography relies on finite fields.

Forcing Transformer attention mechanisms into traditional cryptographic constraints explodes proof generation times from milliseconds to impossible lifetimes.



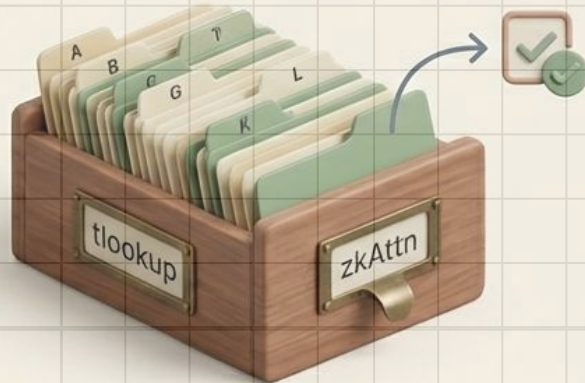


## Old Way



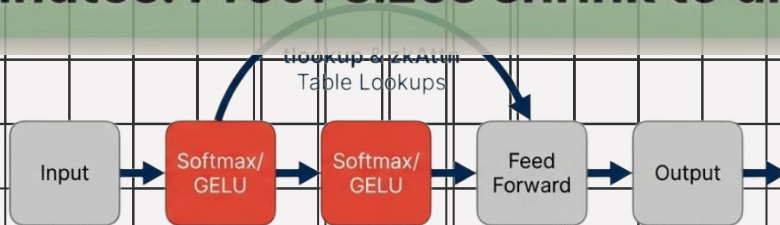
Calculating complex exponential math inside a cryptographic circuit.

## The zkLLM Way



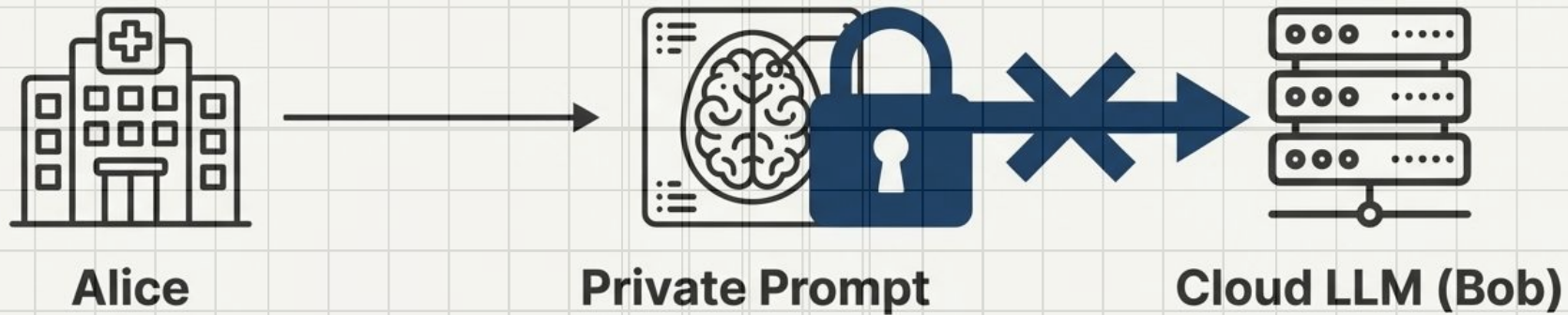
Instantly verifying inputs and outputs against pre-computed 'Lookup Tables' instead of doing the math.

**Result: Proof times for 13-billion parameter models drop to under 15 minutes. Proof sizes shrink to under 200kB.**





# Privacy regulations legally prohibit sending sensitive raw data to the cloud

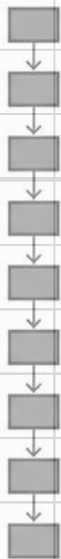


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**The Hospital Scenario:** A medical institution holds top-secret patient medical records. They need an advanced LLM to diagnose a rare disease but legally cannot expose raw HIPAA/GDPR-protected data to a third-party cloud provider.

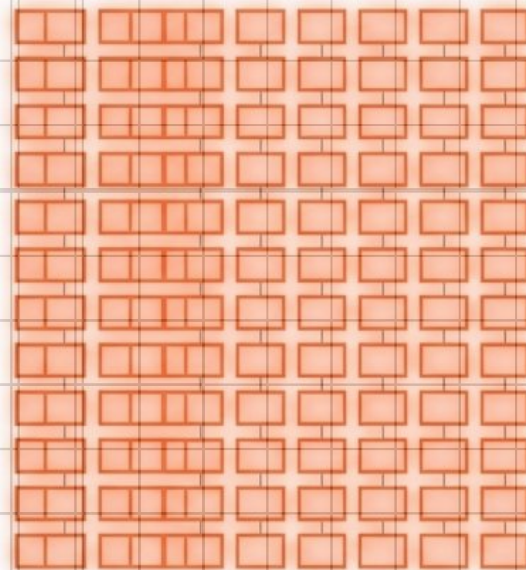
# INNOVATION 1: TLOOKUP

Standard Lookup



Sequential (Slow)

tlookup

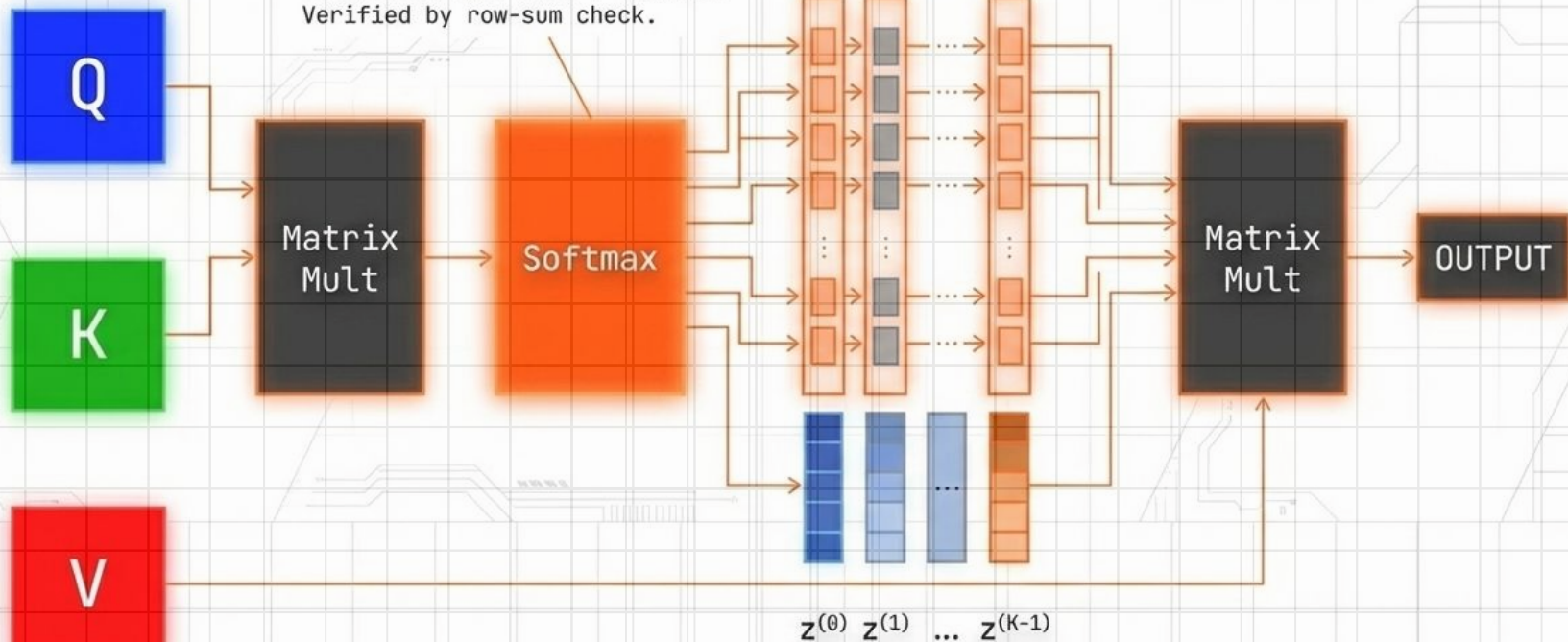


Parallel (GPU Native)

The Breakthrough: Parallelized Lookup Argument  
using Multilinear Extensions

# INNOVATION 2: zkATTN

JetBrains Mono Medium:  
Decomposed into base-b lookups.  
Verified by row-sum check.



Imagine we have a single row in our matrix  $Z$  with three simple values:

$$Z = [1, 2, 3]$$

If we did a **Standard Softmax**, we would calculate the exponents ( $e^1, e^2, e^3$ ):

- $e^1 \approx 2.718$
- $e^2 \approx 7.389$
- $e^3 \approx 20.085$
- **Total Sum**  $\approx 30.192$

Then we divide each by the sum to get our probabilities:

- $2.718/30.192 = \mathbf{0.09}$
- $7.389/30.192 = \mathbf{0.24}$
- $20.085/30.192 = \mathbf{0.67}$

(Notice how  $0.09 + 0.24 + 0.67 = 1.00$ )

$$s(x_i) = \frac{e^{x_i}}{\sum_{j=1}^n e^{x_j}}$$

$$Z' := Z - \hat{z} \mathbf{1}^\top$$

### Step 1: Calculate the shift constant ( $\hat{z}$ )

- $\hat{z} = \ln(\text{Total Sum})$
- $\hat{z} = \ln(30.192) \approx \mathbf{3.407}$

### Step 2: Shift the row ( $Z' := Z - \hat{z} \mathbf{1}^\top$ )

We subtract  $\hat{z}$  from every single element in our original row  $Z$ : ⓘ

- $1 - 3.407 = \mathbf{-2.407}$
- $2 - 3.407 = \mathbf{-1.407}$
- $3 - 3.407 = \mathbf{-0.407}$

So, our new shifted row is  $Z' = [-2.407, -1.407, -0.407]$ .

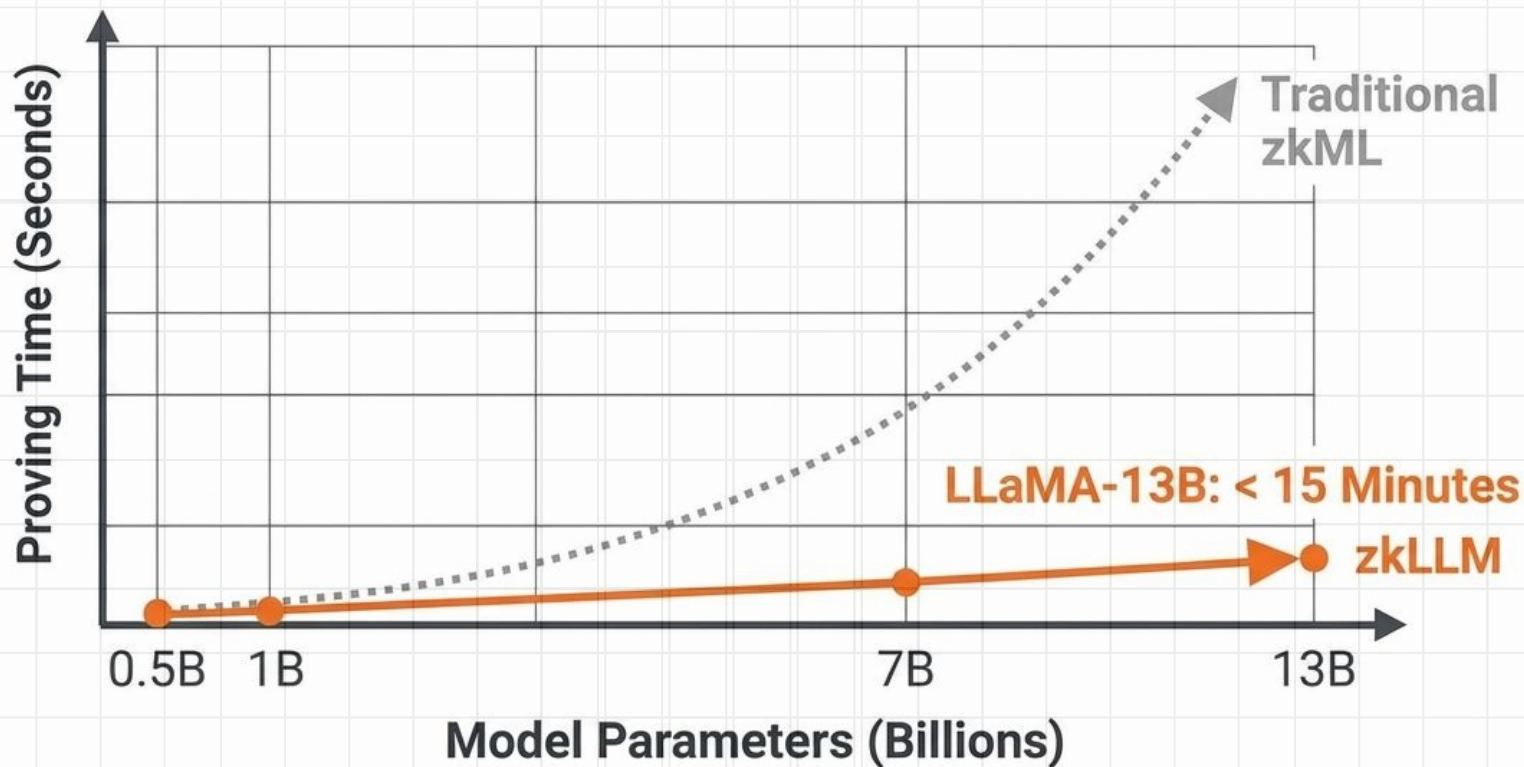
### Step 3: Exponentiate the shifted values

Now, we just apply the exponent to the new shifted values: ⓘ

- $e^{-2.407} \approx \mathbf{0.09}$
- $e^{-1.407} \approx \mathbf{0.24}$
- $e^{-0.407} \approx \mathbf{0.67}$



# THE zkXML SINGULARITY



# FROM THEORY TO INFRASTRUCTURE

Academic Papers



zkLLM



Enterprise API



- **Velocity:** From "Impossible" to "**15 Minutes**" in 2 years.
- **Future:** SLAs for **Correctness**, not just Uptime.



### **The Burden:**

Turning neural network math into massive arithmetic circuits over prime fields requires 986s for committing and 803s for proving.

### **The Silver Lining:**

It takes 13.5 minutes for the cloud to generate the proof, but only 3.95 seconds and minimal memory for the end-user to verify it.

# REFERENCES

- 1. Sun, H., Li, J., & Zhang, H. (2024). zkLLM: Zero Knowledge Proofs for Large Language Models.
- 2. Emergent Mind. (2025). The zkML Singularity.
- 3. BitViraj. The Privacy Trinity: Convergence of MPC, FHE, and Zero Knowledge.
- 4. Boneh et al. (2020). Halo Infinite: Recursive zk-SNARKs.



# “Don’t trust. Verify.”

I will now take your questions.



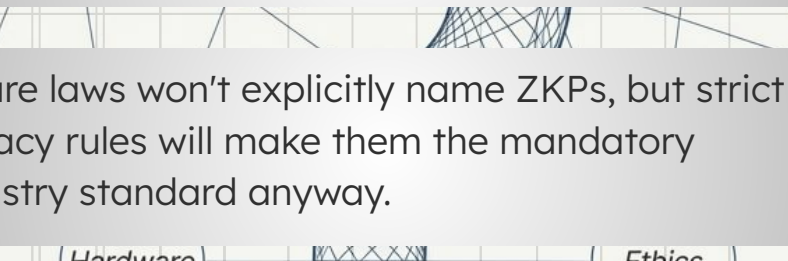
# The Economics and Ethics of Private AI: Let's Debate

**Prompt 1:** If ZK-inference costs 1,000x more compute, who ultimately absorbs the cost? The user demanding privacy, or the cloud provider?

**Prompt 2:** Will future regulations (like the EU AI Act) mandate ZKPs for high-risk AI, forcing the market to adapt regardless of the overhead?



Enterprises will absorb the high compute costs for now. Future hardware advancements will eventually make it cheap enough for everyone else.



Future laws won't explicitly name ZKPs, but strict privacy rules will make them the mandatory industry standard anyway.

## Towards Verifiable AI with Lightweight Cryptographic Proofs of Inference

Pranay Anchuri, Matteo Campanelli (Offchain Labs), Paul Cesaretti (CUNY Graduate Center), Rosario Gennaro (City University of New York and Offchain Labs), Tushar M. Jois, Hasan S. Kayman (City College of New York) and Tugce Ozdemir (CUNY Graduate Center)

 [Abstract](#)